Metals and Non-metals

Science • Chapter 3 • *Chapter Summary & Study Guide for Daksh*

Look around — most useful items in your life are made of **metals** (cars, wires, utensils, coins) or use **non-metals** (oxygen we breathe, the carbon in pencils, sulphur in matches). This chapter compares the physical and chemical behaviour of metals and non-metals, explains how metals are arranged in a **reactivity series**, how they are obtained from their ores (**metallurgy**), and why they corrode.

Physical properties — how they differ

- **Lustre:** metals are shiny when freshly cut; most non-metals are dull (exceptions: iodine, graphite).
- **Hardness:** metals are mostly hard (exceptions Na, K — soft, can be cut with a knife); non-metals vary (diamond is hardest natural substance).
- **Malleability & Ductility:** metals can be hammered into sheets and drawn into wires; non-metals are brittle.
- **Conductivity:** metals conduct heat and electricity (Ag > Cu > Al); non-metals are insulators (graphite excepted).
- **State and m.p./b.p.:** most metals are solids with high m.p. (mercury is liquid; gallium melts in palm); non-metals can be solid, liquid (Br_2) or gases.
- **Sonority:** metals make a ringing sound when struck.

Chemical properties of metals

- **With oxygen:** Metal + $O_2 \rightarrow$ Metal oxide. e.g. $2Cu + O_2 \rightarrow 2CuO$ (black). Most metal oxides are basic; some (Al_2O_3, ZnO) are **amphoteric** — react with both acids and bases.
- **With water:** reactivity decreases as we go down — K, Na react violently with cold water; Mg with hot water; Fe only with steam; Cu, Au don't react.
- **With acids:** Metals above hydrogen displace H_2 from dilute HCl/H_2SO_4. Cu, Ag, Au don't. HNO_3 is a special case — usually it oxidises hydrogen, so H_2 is rarely released.
- **With salt solutions:** a more reactive metal displaces a less reactive one. Iron nail dipped in $CuSO_4$ turns brown; blue colour fades.
- **With non-metals:** Metals lose electrons (form +ve ions); non-metals gain electrons (form –ve ions). e.g. $2Na + Cl_2 \rightarrow 2NaCl$ ($Na^+ + Cl^-$ form an ionic bond).

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REACTIVITY SERIES (decreasing order)
K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au
  most reactive ----------------------> least reactive
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Ionic compounds

When a metal donates electrons and a non-metal accepts them, oppositely charged ions form, held by strong electrostatic forces — these are **ionic compounds**. They are usually solids with high m.p./b.p., dissolve in water but not in petrol, and conduct electricity in **molten or aqueous** form (because ions become mobile).

Metallurgy — how we obtain a metal from its ore

Most metals occur in nature as **ores** (a mineral from which the metal can be extracted profitably). The extraction strategy depends on where the metal lies in the reactivity series:

- **Top of series (K, Na, Ca, Mg, Al)** — extracted by *electrolysis* of their molten salts/oxides.
- **Middle (Zn, Fe, Pb, Cu)** — ore is first concentrated, then converted to oxide (by **roasting** for sulphide ores, **calcination** for carbonate ores), then reduced to the metal usually with *carbon*: e.g. $ZnO + C \rightarrow Zn + CO$.
- **Bottom (Hg, Ag, Au)** — often found native or obtained by simple thermal decomposition of their compounds.
- **Refining:** impure metal is purified by *electrorefining* — impure block as anode, thin pure strip as cathode in a salt-solution electrolyte.

Corrosion and its prevention

Corrosion is the slow chemical attack on metals by air and moisture: iron rusts ($Fe_2O_3 \cdot xH_2O$), silver tarnishes (Ag_2S), copper develops a green coat. Preventing rust means keeping out air or moisture: **painting**, **oiling**, **greasing**, **galvanisation** (zinc coating — zinc sacrifices itself), **chrome plating**, **alloying** (e.g. stainless steel — Fe + Cr + Ni). Alloys often have better properties than pure metals — harder, more resistant to corrosion, with desired colour or melting point. Examples: brass (Cu + Zn), bronze (Cu + Sn), solder (Pb + Sn), 22-carat gold (Au + Cu/Ag).

★ MUST-REMEMBER POINTS

- Metals lose electrons → cations (Na^+); Non-metals gain → anions (Cl^-).
- Activity series: K, Na, Ca, Mg, Al, Zn, Fe, Pb, [H], Cu, Hg, Ag, Au.
- Amphoteric oxides (Al_2O_3, ZnO) react with both acids and bases.
- Roasting (in air) for sulphide ores; Calcination (without air) for carbonate ores.
- Carbon reduction is used for middle-of-series metals (Zn, Fe, Pb, Cu).
- Galvanisation = zinc coating; sacrificial protection. Alloys are homogeneous mixtures.
- Ionic compounds conduct electricity only when molten or in aqueous solution.

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